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Access the code, data, and analysis at https://github.com/jossl95/co-pub_networks

Intersectional Inequalities in Co-Authorship Networks: Gender, Origin, and Collaboration in Dutch Social Science

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1 Introduction

Human knowledge is created, sustained, and spread in an inherently social way. Departmental colleagues, disciplinary conferences, publishing venues and editors, and varieties of academic committees are all central to academic norms and practices (Bourdieu, 1988; Burt, 1998; Stark et al., 2020). Yet simultaneously, academic socialization also constitutes a central pathway through which inequalities emerge and reproduce. For instance, access to prestigious academic positions depend heavily on one’s mentors (Hofstra et al., 2022); dissertation committees exhibit gender-homophilous tendencies in awarding cum laude graduate degrees (Bol, 2023); and the size and composition of co-authorship teams dictate whether papers are disruptive or consolidating in their findings (Wu et al., 2019). Here we argue that paper co-authorships in science are perhaps a key social organization in academe that contributes to and sustains academic inequalities. This is because (a) (peer-reviewed) papers are the essential form to contemporary human knowledge production that (b) is increasingly done in co-authorship teams (REFS) that (c) dynamically shapes productivity and visibility (Abbasi et al., 2011; Ductor, 2015; Li et al., 2022; Moody, 2004; Newman, 2001; Wallace et al., 2012). As such, co-authorship and the resulting output is still the main aspect on which scientists are valued and less access to (more successful) co-authors causes long-lasting negative effects on academic careers.

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Along what axes do co-authorships cause disparities? Particularly problematic ones are divides by gender and ethnic origin: when women, minoritized scholars, or minoritized women scholars find less access to prolific co-authorship, it is one more signal of unequal access to academic careers and it offers narrower epistemological perspectives on our understanding of what good scientific knowledge is. It also signifies reproduced advantage instead of distributed opportunity, and this counters an ideal of a scientific meritocracy (Merton, 1973). We say “one more,” because gender disparities in academic careers are well-documented on many dimensions already (e.g., Barrios et al., 2013; Bol, 2023; Greska, 2023; L. Kim et al., 2022; Lutter & Schröder, 2016): women have lower salaries (C. Kim & Shin, 2024), obtain fewer grants (Stadmark et al., 2020), receive less recognition for often more innovative work (Hofstra et al., 2020), have shorter and less prolific careers (Mulders et al., 2024; Tolsma et al., 2025), and work on topics that yield lower career returns (C. Kim & Shin, 2024). Additionally, minoritized scholars – by national origin, ethnic background, and skin color are less likely to obtain senior positions, have fewer publications and citations, and face slower career progression (Hofstra et al., 2020; Kozlowski et al., 2022; Whittington, 2018).

How do (co-authorship) networks and gender and national origin interrelate in academia? Prior work shows that women are less likely than men to occupy powerful positions in (co-authorship) networks (Gao et al., 2022; Whittington, 2018; Whittington et al., 2024). Women’s collaborations are also more often local and not international (Kwiek & Roszka, 2021), and they populate fewer senior authorship positions (Barrios et al., 2013). Furthermore, scholars with minoritized nationalities to occupy peripheral network positions (Hâncean et al., 2014; Khor & Yu, 2016; J. Kim et al., 2016; Kwiek & Roszka, 2021), limiting exposure to high-impact journals and influential collaborators. Origin-based disadvantages are reinforced by institutional prestige hierarchies (Allison & Long, 1987, 1990; Wapman et al., 2022): work from elite universities and English-speaking or Western countries is more likely to be indexed, cited, and recognised (Khor & Yu, 2016; Li et al., 2022), while research from less visible institutions or non-core regions is underrepresented (Céspedes et al., 2025; Zheng et al., 2025).

Here, we build on earlier insights and examine to what extent network mechanisms – such as homophily, triadic closure, preferential attachment, and brokerage – shape collaboration opportunities and inequalities (Barabasi & Albert, 1999; Bojanowski & Czerniawska, 2020; Ebadi & Schiffauerova, 2015; Gondal, 2011, 2018) and reproduce gender and origin-based segregation in collaboration networks (Gao et al., 2022; Hâncean et al., 2014; Kwiek & Roszka, 2021; Whittington et al., 2024). Therefore, we ask: *to what extent are there gender- and origin-based differences in collaboration (networks)?* and *to what extent are there gender- and origin-based differences in homophily?* *Do actors tend to choose collaborators who share their gender or origin?* We do so by drawing from comprehensive data on a near-population of social scientists in Dutch universities and their co-authorship histories. These time-stamped snapshots function as an anchorpoint in each scholar’s career, fixing their affiliation and co-authorships at a given moment in time and

thereby reducing the influence of survivor and publication bias that typically distort conventional bibliometric network studies (Simko, 2015). This allows us to characterise disparities in collaboration opportunities and outcomes not just for a selected discipline or journal, but for the entire social-science habitat, mirroring recent full-population approaches in the science-of-science literature (Hofstra et al., 2020; Whittington et al., 2024). Prior work interprets gender or origin gaps in collaboration without fully considering network endogenous processes such as transitivity (Gao et al., 2022; Kwiek & Roszka, 2021), possibly overestimating key indicators like homophily. Here, we are among the first to explicitly model path dependency and endogenous network processes such as transitivity by using longitudinal network models. This shifts the focus from descriptive differences to the network processes through which inequalities are reproduced. Because our data provide rich information on all actors in the network and approximate a full-population sample, we can also address survivor bias and sample selection problems that arise in bibliometric studies focused only on highly productive scholars or selected journals (Hâncean et al., 2014; Ossenblok et al., 2014).

We also posit that gender- and origin-based differences in collaboration do not happen in a vacuum and introduce the idea of intersectionality in academic collaboration networks. This provides a framework for understanding how axes of inequality intersect, emphasizing that gender and origin (or race, ethnicity, nationality) are mutually constitutive axes of social stratification (Crenshaw, 1989; King, 1988). This is important, because prior work in science and higher education for instance demonstrates that gender penalties tend to be more pronounced for women from underrepresented nationalities or ethnic groups, while shared nationality may mitigate disadvantages for majority-group women – or vice versa (Kwiek & Roszka, 2021; van der Wal et al., 2021). Approaches that simply consider gender and origin therefore understate the disadvantage that accrues at their intersection and risk missing the specific penalties experienced by intersectional groups (Kozłowski et al., 2022). Therefore, we are among the firsts to explicitly contribute an intersectional perspective to study collaboration opportunities and segregation in collaboration networks among Dutch social scientists. This enables us to ask and answer novel questions, for instance on whether homophily on gender and origin is stronger for majority-group men than for minoritised scholars, whether international women are particularly excluded from triadic closure processes, and whether bridging ties to high-status alters are unequally distributed (Kwiek & Roszka, 2021; Whittington et al., 2024). By focusing on these mechanisms by means of our intersectional perspective, we identify in far more detail than before where in the collaboration process intersectional penalties emerge. Therefore, we ask: *to what extent does gender- and origin-based homophily intersect? Do actors tend to choose collaborators who share both their gender and their origin?*

Finally, we examine the role of social connections in academic networks even further by explicitly studying second order social capital influences. This refers to the resources and opportunities people can access through the contacts of their contacts. This type of capital may be particularly relevant in academic networks because introductions to influential collaborators, invitations to special issues,

or access to well-funded research teams may arise from the networks of mentors, co-authors, or departmental colleagues. Women and minoritized scholars may be less likely to benefit from the indirect networks of their advisors or collaborators. As a result, inequalities compound across network levels, making second-order social capital a key mechanism through which gender, origin, and intersectional disparities in collaboration emerge. To our knowledge, such questions have hardly been asked in the context of co-authorship networks despite reasonable assumptions that such an approach offers us far more insight into how unequal access to high-status, well-connected, and highly productive alters-of-alter shapes cumulative advantage and disadvantage in academic careers. We therefore study to what extent network aspects of second-order social capital - in particular the h-index, degree centrality, and academic position – operate differently across gender and origin groups and their intersection?

i Note

RQ4 nog nader bij te stellen na overleg over welke variabelen we hierin willen benadrukken.
Allen: welke maat vinden jullie hierbij het beste passen?

2 Theory (Outline)

2.1 Gender- and Origin-Based Inequalities in Collaboration Networks

(RQ1: to what extent are there gender- and origin-based differences in collaboration (networks)? — H1a–H1b)

2.1.1 Co-authorship networks as core infrastructure of science

- Present scientific collaboration as a small-world, clustered, and stratified system: short paths, heavy-tailed degree distributions, core–periphery structures.
 - Draw on co-authorship studies across fields and countries (Abbasi et al., 2011; Barabási et al., 2002; Ferligoj et al., 2015; Moody, 2004; Newman, 2001).
- Position co-authorship networks as a central arena where resources and recognition circulate, linking to classic sociology of science and academic stratification.
 - Universalism vs particularism, cumulative advantage, and academic fields as social spaces (Bourdieu, 1988; Long & Fox, 1995; Merton, 1973).

2.1.2 Gendered and ethnic stratification in academic careers

- Summarize persistent gender and ethnic inequalities in access to academic positions, productivity, and recognition.
 - Early stratification and cumulative advantage (Allison & Long, 1987, 1990; Long, 1978; Long et al., 1979; Long & McGinnis, 1981).

- Contemporary evidence on gender and ethnic gaps in publishing, tenure, and careers (Hofstra et al., 2020; Hofstra et al., 2022; Mulders et al., 2024; Tolsma et al., 2025; Wapman et al., 2022).
- Distortions in evaluation and rewards, including cum laude, grants, and media attention (Bol, 2023; Heiberger et al., 2025; Stadmark et al., 2020).

2.1.3 Gendered inequalities in collaboration and network positions

- Argue that women are structurally disadvantaged in collaboration networks: fewer ties, less central positions, and fewer prestigious authorship roles.
 - Empirical evidence on gender disparities in authorships and collaboration patterns (Bendels et al., 2018; Gao et al., 2022; Kwiek & Roszka, 2021; Whittington, 2018; Whittington et al., 2024).
 - Review work on “pipeline leakage” and chilly climates in STEM that shape opportunities to collaborate (Cabay et al., 2018; Dasgupta et al., 2015; Dennehy & Dasgupta, 2017; Greska, 2023).
- Introduce the idea that networks themselves become mechanisms of inequality, not just reflections of individual productivity.
 - Network-based explanations of gender gaps in productivity, prominence, and career success (Collischon & Eberl, 2021; Li et al., 2022; Lutter, 2015; van der Wal et al., 2021; Woehler et al., 2021).

2.1.4 Ethnicity / origin and inequalities in collaboration

- Motivate an origin-based dimension: ethnic and migration background shape opportunities to participate in academic networks and to be recognized.
 - Evidence on ethnic inequalities in academic outcomes (Fuligni et al., 2005; Hofstra et al., 2020; L. Kim et al., 2022; Kozłowski et al., 2022; Mulders et al., 2024).
- Highlight analogies with racialized labor-market networks (job leads, informal support) to argue why minority-origin scholars may have fewer or less advantageous ties.
 - Old-boy networks and racialized access to social capital (Huffman & Torres, 2002; Lalanne & Seabright, 2022; Lutter, 2015; McDonald, 2011; McGuire, 2002).

2.1.5 Co-authorship networks as social capital

- Connect structural position in co-authorship networks to social capital: access to information, collaboration opportunities, and visibility.
 - Structural holes, brokerage, and returns to network position (Burt, 1998; Ebadi & Schiffauerova, 2015; Li et al., 2022; Liu et al., 2025; Yan & Ding, 2009).
- Argue that women and minority-origin scholars have both fewer and differently configured network resources, implying lower degree centrality and less central positions on average.

2.1.6 Hypotheses

- **H1a** On average, women scholars have fewer co-authorship ties, lower degree centrality, and occupy less central network positions than men.
 - **H1b** On average, scholars with a (migration) minority origin have fewer co-authorship ties, lower degree centrality, and occupy less central network positions than scholars with a majority origin.
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2.2 Gender and Origin Homophily in Collaboration Choices

(RQ2: to what extent are there gender- and origin-based differences in homophily? — H2a-H2b)

2.2.1 Homophily as a key mechanism in collaboration

- Introduce the concept of homophily (similarity-based tie formation) as central to understanding co-authorship.
 - General homophily in social and work networks (Huffman & Torres, 2002; Ibarra, 1992; Ibarra, 1995; McDonald, 2011; L. Zhang, 2017).
- In scientific collaboration, homophily interacts with transitivity and preferential attachment.
 - ERGM-based and dynamic analyses of collaboration formation (Gondal, 2011; Leifeld, 2018; Purwitasari et al., 2020; Stark et al., 2020; C. Zhang et al., 2018).

2.2.2 Gender homophily in collaboration networks

- Summarize evidence that co-authorship often follows gender lines, though constrained by male-majority contexts.
 - Gendered patterns of collaboration, prestige positions, and team roles (Gao et al., 2022; Kretschmer et al., 2015; Kwiek & Roszka, 2021; Whittington, 2018; Whittington et al., 2024).
 - Studies of female peers, majority-female groups, and mentors enhancing women’s participation and persistence (Dasgupta et al., 2015; Dennehy & Dasgupta, 2017; Hofstra et al., 2022).
- Combine this with work on gendered networks and returns to argue why we expect gender homophily in co-authorship choices (H2a), with implications for inequality (Lalanne & Seabright, 2022; Lutter, 2015; Woehler et al., 2021).

2.2.3 Origin / nationality homophily

- Present national, ethnic, and origin-based homophily as a robust mechanism in friendship and collaboration networks.

- Empirical evidence on national / ethnic homophily (Hâncean et al., 2014; Ibarra, 1995; Pillai et al., 2019; Santos et al., 2024).
- Studies of departmental and field-level specializations and exchanges (which often map onto national and origin-based lines) (Abbasi et al., 2011; Ferligoj et al., 2015; Gondal, 2018; Heiberger & Riebling, 2019).
- Argue that shared origin can lower barriers of trust, language, and informal communication, motivating H2b (higher likelihood of co-authorship among same-origin scholars).

2.2.4 Homophily, polarization, and the reproduction of inequality

- Link homophily to segmentation and polarization in scientific collaboration.
 - Polarization and assortative mixing in social science co-authorship (Hâncean et al., 2014; Leifeld, 2018; Moody, 2004).
- Argue that gender and origin homophily, operating in a structurally unequal field, can reinforce existing disparities in centrality and access to high-status collaborators.

2.2.5 Hypotheses

- **H2a** Co-authorship ties are more likely between scholars who share the same gender than between scholars of different genders.
- **H2b** Co-authorship ties are more likely between scholars who share the same origin than between scholars with different origins.

2.3 Intersectional Homophily: Joint Patterns of Gender and Origin

(RQ3: to what extent does gender- and origin-based homophily intersect? — H3a–H3c)

2.3.1 Intersectionality in science and knowledge production

- Introduce an intersectional perspective: inequalities in science are not simply additive by gender and origin, but emerge at their intersections.
 - Large-scale evidence on intersectional inequalities in science and knowledge production (Hofstra et al., 2020; L. Kim et al., 2022; Kozłowski et al., 2022; Mulders et al., 2024).
- Emphasize that group positions (e.g., majority-origin men vs minority-origin women) differ in both the opportunities to collaborate and the payoffs to collaboration.

2.3.2 Intersectional patterns in collaboration networks

- Discuss how network-based inequalities can be intersectional: some intersections occupy dense, high-status cores; others are more peripheral.
 - Work on stratification in collaboration networks and status hierarchies (Ferligoj et al., 2015; Heiberger & Riebling, 2019; Moody, 2004; Stark et al., 2020; Wapman et al., 2022).

- Evidence that collaboration structures and returns differ across gender and regions/fields (Li et al., 2022; van der Wal et al., 2021; Whittington, 2018; Whittington et al., 2024).
- Argue that majority-origin men are overrepresented in central, well-connected cores, making same-gender and same-origin collaboration both easier and more rewarding.

2.3.3 Constraints and strategies of minority-origin women

- Draw on research suggesting that women and ethnic minorities face structural constraints that reduce their ability to rely exclusively on homophilous networks.
 - Old-boy networks, lower-status contacts, and constrained access to high-quality ties (Huffman & Torres, 2002; Lutter, 2015; McDonald, 2011; McGuire, 2002; Woehler et al., 2021).
 - Intersectional disadvantages in science and academic careers (Hofstra et al., 2020; Kozlowski et al., 2022; Mulders et al., 2024).
- Combine this with work showing that minority and women actors often develop mixed or bridging collaboration strategies to navigate exclusionary structures.
 - Women’s networking strategies and gendered returns in executive and digital networks (Contreras et al., 2025; Lalanne & Seabright, 2022; Woehler et al., 2021).
- From this, reason that minority-origin women may show weaker intersectional homophily than majority-origin men, because they must diversify collaborators across gender and origin to maintain careers (H3c), even while still facing subtle penalties.

2.3.4 Intersectional homophily as a mechanism of cumulative advantage

- Link back to cumulative advantage and status closure: homophily along intersecting lines of gender and origin can magnify existing differences in degree centrality, h-index evolution, and access to high-prestige venues.
 - Cumulative advantage and status closure in academia (Hofstra et al., 2020; Long & Fox, 1995; Merton, 1973; Simko, 2015; Wapman et al., 2022).
- This motivates modeling joint gender–origin configurations rather than separate effects.

2.3.5 Hypotheses

- **H3a** Co-authorship ties are especially likely between scholars who share both gender and origin (e.g., majority-origin men with majority-origin men; minority-origin women with minority-origin women), over and above the separate effects of gender and origin homophily.
- **H3b** Majority-origin men exhibit the highest levels of intersectional homophily (preferring other majority-origin men), reinforcing their dominance in central parts of the network.
- **H3c** Minority-origin women show weaker intersectional homophily than majority-origin men, because structural constraints and limited opportunities push them into more mixed-gender and mixed-origin collaborations.

2.4 Second-Order Social Capital and Unequal Returns Across Gender–Origin Intersections

(RQ4: to what extent do network aspects of second-order social capital (h-index, degree centrality, academic position) operate differently across gender–origin intersections? — H4a–H4b)

2.4.1 Conceptualizing second-order social capital

- Define second-order social capital as the quality and embeddedness of an ego’s collaborators: collaborators’ productivity, impact, and positions (e.g., their h-indices, centralities, academic ranks).
 - h-index as a key indicator of an individual’s cumulative impact (Hirsch, 2005).
 - Impact of co-author networks, centrality, and structural position on scientific productivity and citations (Ductor, 2015; Ebadi & Schiffauerova, 2015; Liu et al., 2025; Wu et al., 2019; Yan & Ding, 2009).
- Connect to the social capital and structural holes literatures: who you are connected to (and who they are connected to) shapes access to resources.
 - Structural holes, brokerage, and returns (Bojanowski & Czerniawska, 2020; Burt, 1998; Collischon & Eberl, 2021; Mahmood et al., 2021).

2.4.2 Academic position and departmental context as network resources

- Situate academic position (PhD, postdoc, assistant, associate, full professor) as both a marker of status and a generator of collaboration opportunities.
 - Classic work on productivity, position, and departmental context (Allison & Long, 1990; Long, 1978; Long et al., 1979; Long & Fox, 1995; Long & McGinnis, 1981).
 - Contemporary evidence on tenure and professorship in sociology (Lutter & Schröder, 2016).
- Link academic rank to visibility and attractiveness as a collaborator: higher positions typically have more collaborators and more central roles in networks.

2.4.3 How second-order social capital affects tie formation

- Argue that scholars with higher second-order social capital (high-impact, central, high-rank collaborators) become more attractive collaboration partners, increasing their chances of gaining new ties (H4a).
 - Preferential attachment and cumulative advantage in collaboration (Abbasi et al., 2011; Baraba et al., 2002; Ferligoj et al., 2015; J. Kim et al., 2016; C. Zhang et al., 2018).
 - Project-team and network studies linking structural properties to impact and innovation (Liu et al., 2025; Lungeanu & Contractor, 2015).

2.4.4 Unequal returns to networks: gender and origin differences

- Introduce the unequal network returns (UNR) perspective: even when women and minority scholars have similar network structures, they may receive lower payoffs in terms of career outcomes.
 - Framework and review of UNC vs UNR (Woehler et al., 2021).
 - Evidence of gendered returns to networks and professional ties (Contreras et al., 2025; Lalanne & Seabright, 2022; Lutter, 2015; Whittington, 2018; Whittington et al., 2024).
 - Labor-market analogues showing gendered and racialized differences in job leads and returns to contacts (Huffman & Torres, 2002; Lutter, 2015; McDonald, 2011; McGuire, 2002).
- Connect to research on gendered and intersectional returns to productivity and innovation:
 - Diversity–innovation paradox and devaluation of minority innovations (Hofstra et al., 2020; Kozlowski et al., 2022).
 - Gender disparities in high-impact publication, grant success, and recognition (Bendels et al., 2018; Heiberger et al., 2025; Stadmark et al., 2020; Tolsma et al., 2025).

2.4.5 Intersectional differences in second-order returns

- Argue that majority-origin men are most likely to be embedded in high-status, high-impact collaboration neighborhoods, and that evaluators interpret their networks more favorably.
 - Hierarchy and prestige in hiring and retention (Heiberger & Riebling, 2019; Wapman et al., 2022).
 - Gendered knowledge and topic valuation (Hofstra et al., 2020; L. Kim et al., 2022; Kozlowski et al., 2022).
- Women and minority-origin scholars may have less access to high second-order capital and/or receive lower marginal returns to similar second-order profiles, because their collaborators' prestige is discounted or because evaluators interpret the same network signals differently.

2.4.6 Hypotheses

- **H4a** Higher levels of second-order social capital (higher h-index, higher degree centrality, and higher academic position) are associated with a greater likelihood of attracting new co-authorship ties for all scholars.
- **H4b** The positive association between second-order social capital and attracting new co-authorship ties is stronger for men and majority-origin scholars than for women and minority-origin scholars.

3 Data and Methods

3.1 Data

This study contributes a database of $N_{\text{scholars}} = \text{XXXX}$ scholars and their $N_{\text{publications}} = \text{XXXX}$ publications that is near-population of staff employed in Sociology ($N_{\text{sociology}} = \text{XXXX}$ unique academics) and Political Science ($N_{\text{politicalscience}} = \text{XXXX}$ unique academics) departments at Dutch research universities in recent years and their academic histories. We compiled these data by reconstructing rosters of department employees at three time points - 19 December 2022, 19 April 2024, and 12 October 2025 - by manually collecting staff lists from public departmental webpages of these research universities. The resulting database contains for each of the three time points staff members' their names, university affiliations, job titles, and email addresses

To be able to consider co-authorship networks, we needed to link these scholars to publication histories. To this end, we use OpenAlex as a source that is increasingly used in bibliometric research (Priem et al., 2022). OpenAlex is open source and contains scholarly metadata. It offers publication coverage comparable to major proprietary databases such as Web of Science or Scopus, with a particularly strong coverage of persistent identifiers that enable reliable author disambiguation (e.g. ORCID so as to pinpoint which authors wrote what publications). It also provides a more balanced linguistic coverage than Web of Science, albeit with some remaining limitations in the accuracy and completeness of language metadata (see, e.g., Céspedes et al., 2025; Culbert et al., 2025).

How do we link departmental scholars to OpenAlex? We first queried OpenAlex for institutions (both their English and Dutch names) to find OpenAlex institution identifiers, such that we can match scholars' names and their OpenAlex institutional identifiers. In a next step, we query OpenAlex for full name + OpenAlex institutions to be able to link our scholars to OpenAlex scholar identifiers. Note that this crosslinking procedure allows for each scholar to be linked to multiple OpenAlex author IDs. This is important for our purposes – and simultaneously highlights the importance of having a relevant scholarly anchor database – because some scholars have more than one profile (XX% of our scholars) in OpenAlex due to changing names or institutional affiliations across publications. So we now have candidate OpenAlex scholar identifier and profiles that we filter through institutional matches, name similarity thresholds (Jaro–Winkler similarity), and a small set of manual heuristics. Our aim is to systematically exclude false matches and retain a high-confidence links between our scholars and OpenAlex scholar identifiers. We then compile for each linked scholar and their associated OpenAlex scholar identifier their publications, such that each row represents a unique author-publication. The associated publication metadata consists of information like OpenAlex publication identifiers, publications years, what type of publication it is, and, most importantly for our purposes, with whom the publication is written. XX% of our

scholars were matched to publication data. All code and (intermediate) output of this study are documented and made available through our public replication package website: [XXURLXXX](#).

3.2 Variables

3.2.1 Dependent Variable

Collaboration, the dependent variable, was operationalized as a binary co-publishing tie between pairs of scholars. Using the OpenAlex-based publication data described above, we identified all works in which any two scholars in our roster appeared together in the authorship list. For each pair of scholars, we then coded a tie as present (1) if they had coauthored at least one publication within the observation window, and absent (0) otherwise. This information was assembled into a series of adjacency matrices in which nodes represent scholars and undirected edges represent observed co-publishing relationships. These longitudinal collaboration networks form the outcome in our models, capturing both the existence and evolution of co-authorship ties over time.

Note

- dichotomizing the right way? what do we loose if we do so?
- 1 OR: from first author to all other authors?
- 2 OR: from last author to all other authors?
- 3 OR: a bipartite network of people in a paper that are then linked through that higher order entity? Creating more variation in this way?
- 4 OR: a giant ERGM just looking at it as a single network?
- So we should think about: what are the underlying data generating processes that are most important to answering our research question. Perhaps we also need to 1 and 2 at the very elast and show how differents assumptions on how collaborating starts does not really matter for the main results, though if it does we should intuited why.
- As to 3, we cast the problematization of collaboration in an entirely different way, as the higher order entity (papers) also has a topic, time, length, outlet, potential to impact, etc. It does show us how topical foci perhaps drive some of the collaboration.
- tldr: Important decisions that we should all agree on.

More practical:

- are these isolated communities WITHIN departments? or are links between departments allowed? which one is fair as to our research question?
- and are links allowed between political science and sociology? or do we analyze them as separate networks? what is the consequence of this decision?

That we basically expand the opportunity set with many resulting nonlinks if we allow them to crosslink. Do we want that?

3.2.2 Independent Variables

Gender was operationalized as an inferred gender indicator based on scholars' first names. We first used the `genderizeR` package to query the `genderize.io` database, which provides probabilistic gender assignments for first names based on large-scale social media data (Wais, 2016, see also `genderize.io` documentation). The underlying database aggregates first-name and gender information for more than one hundred million individuals across a wide range of countries, and returns both a predicted gender and associated probability for each name. Using this API, we obtained an initial classification for all unique first names in our scholar roster. To improve coverage for names that are common in the Netherlands and reduce misclassification, we then matched first names to the Dutch public civic registry name database hosted by the Meertens Instituut (Nederlandse Voornamenbank), which contains all officially registered first names in the Netherlands and their distribution by sex over time (Meertens Instituut, n.d.). Where available, this registry information was used to validate or refine the `genderize.io`-based assignments, especially for names with low counts or ambiguous probabilities in `genderize.io`.

Origin was operationalized as an inferred country-of-origin indicator based on scholars' surnames. We first used the `Demografixer` package to query the `nationalize.io` API, which provides probabilistic nationality assignments for last names based on large-scale web data (Brenninkmeijer, 2025, see also `nationalize.io` documentation). The underlying database aggregates name-country occurrences from diverse online sources, and returns, for each surname, a set of country codes with associated probabilities. Using this API, we obtained an initial nationality distribution for all unique surnames in our scholar roster. To improve coverage for surnames that are common in the Netherlands and to correct systematic misclassifications, we then complemented these predictions with information from the Dutch Database of Surnames in the Netherlands (Nederlandse Familienamenbank), which reports the number of name bearers per surname and provides etymological notes on its origin and meaning (Centrum voor familiegeschiedenis, n.d.). A custom scraping and parsing pipeline extracted origin-related signals from these etymology texts (e.g., language adjectives such as "Nederlandse", "Turkse", or "Marokkaanse", references to specific countries or regions, and indicators of Dutch surname particles), and combined them with national occurrence counts. And we further harmonized the various signals to ISO 3166-1 alpha-2 country codes. The resulting variable captures, for each scholar, the most likely country-of-origin signal implied by their surname. In a final than further aggregated origin, to identify whether people are from the following backgrounds: (1) Dutch; (2) Turkish, Surinamese, Moroccan, Dutch Antillean or (3) other.

Position was operationalized as an indicator of scholars' academic rank/role within the Dutch university system. Starting from the free-text job titles collected from departmental staff pages, we

first cleaned and standardized these titles (e.g., “universitair docent”, “universitair hoofddocent”, “hoogleraar”, “postdoc-onderzoeker”, “promovendus”) and translated them into a common English terminology. We then mapped all titles to a set of seven analytically meaningful categories: *PhD student*, *Postdoc*, *Lecturer*, *Researcher*, *Assistant professor*, *Associate professor*, and *Full professor*. In cases where multiple roles were listed for the same individual (e.g., combined teaching and research appointments or endowed/visiting positions), we applied a simple priority hierarchy that privileges structural career rank (professor > associate professor > assistant professor > researcher or lecturer > lecturer > PhD student). The resulting harmonized variable therefore captures each scholar’s primary academic position.

h-index was used as a bibliometric indicator of scholars’ cumulative research impact. For each scholar, we drew on the OpenAlex publication and citation data linked to their author identifiers and calculated the Hirsch index, defined as the largest integer h such that the scholar has at least h publications with $\geq h$ citations each (Hirsch, 2005). We first assembled, for every scholar, the full set of works retrieved from OpenAlex, along with their citation counts at the time of data collection. These works were then sorted in descending order of citations, and the h-index was computed by identifying the point at which the number of publications equals or exceeds the citation threshold. The resulting measure reflects each scholar’s career-wide citation impact within the coverage of OpenAlex over the observation window and is used as a continuous indicator of scholarly standing in our analyses.

3.2.3 Control Variables

Topic Similarity controls for topical proximity between authors two years prior each start of each wave, we constructed a dyadic dissimilarity measure based on OpenAlex topic assignments. For each author (i), we first extracted all publications in a two-year retrospective window $[t - 2, t)$ and created an author–topic profile as the vector of topic shares (row-normalized topic counts). Next, we estimated a topic *proximity* matrix from historical publication data (a broader retrospective window) using cosine similarity between topics; proximities were rescaled to $[0, 1]$ and converted to distances $M = 1 - \text{proximity}$. Author-pair dissimilarity D_{ij} was then computed as a weighted distance between their topic-share profiles, where differences in topic shares are combined across all topic pairs and weighted by topic distances M . Finally, D_{ij} was normalized by the maximum observed value within each year $\times$ topic type so that higher values indicate greater topical dissimilarity (i.e., lower topic similarity) between authors i and j . This dyadic covariate is included as a control in network models to account for homophily in collaboration driven by shared research interests.

Interdisciplinarity controls for the diversity of topics of each author. We operationalized an author’s interdisciplinarity as the extent to which their publication portfolio spans *distant* rather than closely related topics. For each wave t and topic type (OpenAlex subfields), we first constructed an author–topic profile using all publications in a two-year retrospective window $[t - 2, t)$. Topic counts were row-normalized to obtain a vector of topic shares P_i for each author i . We then derived a topic

proximity matrix from historical publication data using cosine similarity between topics, rescaled to $[0, 1]$, such that higher values indicate greater topical relatedness. Interdisciplinarity was computed as one minus the proximity-weighted concentration of an author’s portfolio, $ID_i = 1 - P_i^\top \text{prox } P_i$. This measure increases when an author distributes their work across less-related topics and decreases when publications are concentrated within closely connected topics. Scores were calculated separately for each year and topic type and used as an author-level control to account for individual differences in breadth of scholarly activity that may shape collaboration opportunities and partner choice.

Affiliation captures scholars’ primary institutional location and is measured as a categorical indicator of the Dutch research university at which they are employed. Based on the initial manual scrape of departmental staff pages, we distinguish eight universities: Radboud University, Erasmus University Rotterdam, University of Amsterdam, Leiden University, Vrije Universiteit Amsterdam, Utrecht University, Tilburg University, and University of Groningen. We include affiliation as a control variable to account for structural differences between universities in size, funding, and collaboration culture that may shape co-publishing opportunities.

To account for collaboration opportunities arising from spatial proximity, we included a dyadic control capturing the estimated travel time between authors’ institutional locations. For each year t , we assigned each author to the university of their *first recorded affiliation* in that year. Using these yearly university assignments, we constructed an $N \times N$ matrix where each entry represents the travel time (in hours) between the universities of author i and author j . The travel-time by train in hours was sourced from the 9292 website, which is the main public transport API for the Netherlands. Author pairs with missing affiliation information in year t receive missing values for this dyadic covariate. This variable is included as a control in network models to adjust for the fact that geographically closer scholars may be more likely to collaborate due to lower coordination costs and greater opportunities for face-to-face interaction.

Discipline indicates whether a scholar is employed in sociology or political science, derived from the manually coded departmental information in the initial staff rosters and coded as a binary indicator for sociology versus political science. We include discipline as a control variable to capture field-specific norms in publishing, collaboration rates, and typical team sizes that could otherwise confound observed differences in network positions and outcomes.

i Note

See `cat_desc` for a summary statistics table

3.3 Analytical Approach

Tip

@ jochem zouden wij hierover kunnen praten en samen het werk verdelen voor het beschrijven van de analytische strategie

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